Annex to *Tides in Kaula's formalism for an N-body code*: exact coefficients of the eccentricity functions $G_{2pq}(e)$

Content. The main document prints the series of $G_{20q}(e)$ and $G_{21q}(e)$ (Tables C.3 and C.4) with long fractions rounded to 8 significant digits. This annex gives, for *exactly the same terms*, the exact rational coefficients c_n of

$$G_{2pq}(e) = \sum_n c_n \, e^n,$$

together with their value in double precision.

Origin. The coefficients are obtained from Tisserand's formula (Kaula 2000, eq. 3.67–3.69), expanded in exact rational arithmetic; no floating-point operation is involved. They are validated against the direct numerical Hansen integral to machine precision.

Symmetries. $G_{22q} = G_{20,-q}$ and $G_{21q} = G_{21,-q}$; only G_{20q} ($|q| \le 24$) and G_{21q} ($q \le 0$) are listed. $G_{2pq} = \mathcal{O}(e^{|q|})$, and only every other power of e appears.

Complete series. The full series, to e^{40} and for $|q| \le 30$, are in

- `data/G_exact_l2.txt`: plain text, one line `l p q n c_n` per coefficient, exact fractions;
- `data/G_series.json`: the same, machine-readable (also $l = 3$).

Table 1: Exact coefficients c_n of $G_{20q}(e) = \sum_n c_n e^n$ (terms printed in Table C.3 of the main document).

q	n	c_n (exact)	c_n (decimal)
-24	24	$\frac{81402749386839761113321}{5127672741597020160000}$	1.587518421885223e + 01
	26	$-\frac{14082675643923278672604533}{256383637079851008000000}$	−5.492813739722872e + 01
	28	$\frac{26179938230301535571655166 81}{26663898256304504832000000}$	9.818496147355896e + 01
-23	23	$\frac{38164393614122297481 0969}{32121729178906132480000}$	1.188117657102479e + 01
	25	$-\frac{39309325422545966405529807}{10278953337249962393600 00}$	−3.824253708798605e + 01
	27	$\frac{13193812516338219462190009299}{20557906674499924787200 0000}$	6.417877425576533e + 01
-22	22	$\frac{30517578125000}{3430178002251}$	8.896791392450574e + 00
	24	$-\frac{2090454101562500}{78894094051773}$	−2.649696566882018e + 01
	26	$\frac{19702911376953125}{473364564310638}$	4.162312277355726e + 01
-21	21	$\frac{37589973457545958193355601}{56392353451202523955200 00}$	6.665792639789958e + 00
	23	$-\frac{90591836032685759245986998 41}{49625271037058221080576 0000}$	−1.825518211578840e + 01
	25	$\frac{48870724492155500247181616 8601}{18262099741637425357651968000}$	2.676073681753620e + 01

q	n	c_n (exact)	c_n (decimal)
-20	20	$\frac{1853020188851841}{370812682240000}$	$4.997186659469527\mathrm{e}+00$
	22	$-\frac{1853020188851841}{148325072896000}$	$-1.249296664867382\mathrm{e}+01$
	24	$\frac{7784537813366584041}{456841224519680000}$	$1.703991976983075\mathrm{e}+01$
-19	19	$\frac{14063084452067724991009}{3751592141361512448000}$	$3.748564322070471\mathrm{e}+00$
	21	$-\frac{2545418285824258223372629}{300127371308920995840000}$	$-8.481126778684441\mathrm{e}+00$
	23	$\frac{4920673249778496974354049 1}{4583763489081702481920000}$	$1.073500685953648\mathrm{e}+01$
-18	18	$\frac{274877906944}{97692469875}$	$2.813706187341903\mathrm{e}+00$
	20	$-\frac{137438953472}{24105934125}$	$-5.701457274350699\mathrm{e}+00$
	22	$\frac{62036507623424}{9280784638125}$	$6.684403317428693\mathrm{e}+00$
-17	17	$\frac{1081219482421875}{511612209332224}$	$2.113357466259698\mathrm{e}+00$
	19	$-\frac{15497479248046875}{4092897674657792}$	$-3.786432127048625\mathrm{e}+00$
	21	$\frac{232462188720703125}{56556404231634944}$	$4.110271716861995\mathrm{e}+00$
-16	16	$\frac{678223072849}{426995712000}$	$1.588360383462118\mathrm{e}+00$
	18	$-\frac{35945822860997}{14517854208000}$	$-2.475973538926243\mathrm{e}+00$
	20	$\frac{521553543020881}{209057100595200}$	$2.494789899678040\mathrm{e}+00$
-15	15	$\frac{3937376385699289}{3296144130048000}$	$1.194540114252210\mathrm{e}+00$
	17	$-\frac{66935398556887913}{42190644864614400}$	$-1.586498589241216\mathrm{e}+00$
	19	$\frac{2523858263233244249}{1687625794584576000}$	$1.495508228975911\mathrm{e}+00$
-14	14	$\frac{157464}{175175}$	$8.988953903239617\mathrm{e}-01$
	16	$-\frac{78732}{79625}$	$-9.887849293563580\mathrm{e}-01$
	18	$\frac{1554957}{1751750}$	$8.876591979449122\mathrm{e}-01$
-13	13	$\frac{3138428376721}{4637432217600}$	$6.767599459049827\mathrm{e}-01$
	15	$-\frac{3138428376721}{5299922534400}$	$-5.921649526668599\mathrm{e}-01$
	17	$\frac{1258509779065121}{2397195730944000}$	$5.249924997027789\mathrm{e}-01$
-12	12	$\frac{9765625}{19160064}$	$5.096864498991235\mathrm{e}-01$
	14	$-\frac{166015625}{498161664}$	$-3.332565249340423\mathrm{e}-01$
	16	$\frac{8740234375}{27897053184}$	$3.133031405629915\mathrm{e}-01$
-11	11	$\frac{387420489}{1009254400}$	$3.838680207884157\mathrm{e}-01$
	13	$-\frac{387420489}{2306867200}$	$-1.679422590949319\mathrm{e}-01$
	15	$\frac{64699221663}{335879864320}$	$1.926260801432254\mathrm{e}-01$
-10	10	$\frac{4096}{14175}$	$2.889594356261023\mathrm{e}-01$
	12	$-\frac{2048}{31185}$	$-6.567259900593234\mathrm{e}-02$

q	n	c_n (exact)	c_n (decimal)
	14	$\frac{58624}{467775}$	$1.253252097696542\mathrm{e}-01$
-9	9	$\frac{5764801}{26542080}$	$2.171947714723187\mathrm{e}-01$
	11	$-\frac{5764801}{1061683200}$	$-5.429869286807967\mathrm{e}-03$
	13	$\frac{8237900629}{93428121600}$	$8.817367285055210\mathrm{e}-02$
-8	8	$\frac{729}{4480}$	$1.627232142857143\mathrm{e}-01$
	10	$\frac{243}{8960}$	$2.712053571428572\mathrm{e}-02$
	12	$\frac{24057}{358400}$	$6.712332589285715\mathrm{e}-02$
-7	7	$\frac{15625}{129024}$	$1.211015004960317\mathrm{e}-01$
	9	$\frac{171875}{4128768}$	$4.162864079551092\mathrm{e}-02$
	11	$\frac{16015625}{297271296}$	$5.387545052449329\mathrm{e}-02$
-6	6	$\frac{4}{45}$	$8.888888888888889\mathrm{e}-02$
	8	$\frac{2}{45}$	$4.444444444444445\mathrm{e}-02$
	10	$\frac{11}{252}$	$4.365079365079365\mathrm{e}-02$
-5	5	$\frac{81}{1280}$	$6.328125000000000\mathrm{e}-02$
	7	$\frac{81}{2048}$	$3.955078125000000\mathrm{e}-02$
	9	$\frac{19359}{573440}$	$3.375941685267857\mathrm{e}-02$
-4	4	$\frac{1}{24}$	$4.166666666666666\mathrm{e}-02$
	6	$\frac{7}{240}$	$2.916666666666667\mathrm{e}-02$
	8	$\frac{131}{5760}$	$2.274305555555555\mathrm{e}-02$
	10	$\frac{4303}{241920}$	$1.778687169312169\mathrm{e}-02$
-3	3	$\frac{1}{48}$	$2.083333333333333\mathrm{e}-02$
	5	$\frac{11}{768}$	$1.432291666666667\mathrm{e}-02$
	7	$\frac{313}{30720}$	$1.018880208333333\mathrm{e}-02$
	9	$\frac{3355}{442368}$	$7.584183304398148\mathrm{e}-03$
-2	–	0	0
-1	1	$-\frac{1}{2}$	$-5.000000000000000\mathrm{e}-01$
	3	$\frac{1}{16}$	$6.250000000000000\mathrm{e}-02$
	5	$-\frac{5}{384}$	$-1.302083333333333\mathrm{e}-02$
	7	$-\frac{143}{18432}$	$-7.758246527777778\mathrm{e}-03$
	9	$-\frac{9097}{1474560}$	$-6.169297960069445\mathrm{e}-03$
0	0	1	$1.000000000000000\mathrm{e}+00$
	2	$-\frac{5}{2}$	$-2.500000000000000\mathrm{e}+00$

q	n	c_n (exact)	c_n (decimal)
	4	$\frac{13}{16}$	$8.125000000000000\mathrm{e}-01$
	6	$-\frac{35}{288}$	$-1.215277777777778\mathrm{e}-01$
	8	$-\frac{5}{576}$	$-8.680555555555556\mathrm{e}-03$
	10	$-\frac{49}{3600}$	$-1.361111111111111\mathrm{e}-02$
1	1	$\frac{7}{2}$	$3.500000000000000\mathrm{e}+00$
	3	$-\frac{123}{16}$	$-7.687500000000000\mathrm{e}+00$
	5	$\frac{489}{128}$	$3.820312500000000\mathrm{e}+00$
	7	$-\frac{1763}{2048}$	$-8.608398437500000\mathrm{e}-01$
	9	$\frac{13527}{163840}$	$8.256225585937500\mathrm{e}-02$
2	2	$\frac{17}{2}$	$8.500000000000000\mathrm{e}+00$
	4	$-\frac{115}{6}$	$-1.916666666666667\mathrm{e}+01$
	6	$\frac{601}{48}$	$1.252083333333333\mathrm{e}+01$
	8	$-\frac{1423}{360}$	$-3.952777777777778\mathrm{e}+00$
	10	$\frac{48619}{69120}$	$7.033998842592593\mathrm{e}-01$
3	3	$\frac{845}{48}$	$1.760416666666667\mathrm{e}+01$
	5	$-\frac{32525}{768}$	$-4.235026041666666\mathrm{e}+01$
	7	$\frac{208225}{6144}$	$3.389078776041666\mathrm{e}+01$
	9	$-\frac{6122725}{442368}$	$-1.384079544632523\mathrm{e}+01$
4	4	$\frac{533}{16}$	$3.331250000000000\mathrm{e}+01$
	6	$-\frac{13827}{160}$	$-8.641875000000000\mathrm{e}+01$
	8	$\frac{104127}{1280}$	$8.134921875000001\mathrm{e}+01$
	10	$-\frac{146017}{3584}$	$-4.074135044642857\mathrm{e}+01$
5	5	$\frac{228347}{3840}$	$5.946536458333333\mathrm{e}+01$
	7	$-\frac{3071075}{18432}$	$-1.666164822048611\mathrm{e}+02$
	9	$\frac{132485857}{737280}$	$1.796954440646701\mathrm{e}+02$
6	6	$\frac{73369}{720}$	$1.019013888888889\mathrm{e}+02$
	8	$-\frac{775727}{2520}$	$-3.078281746031746\mathrm{e}+02$
	10	$\frac{7527241}{20160}$	$3.733750496031746\mathrm{e}+02$
7	7	$\frac{12144273}{71680}$	$1.694234514508929\mathrm{e}+02$
	9	$-\frac{252361629}{458752}$	$-5.501046949114118\mathrm{e}+02$
	11	$\frac{13581249561}{18350080}$	$7.401193652016776\mathrm{e}+02$
8	8	$\frac{634085}{2304}$	$2.752105034722222\mathrm{e}+02$

q	n	c_n (exact)	c_n (decimal)
	10	$-\frac{277861775}{290304}$	$-9.571407042272928\mathrm{e}+02$
	12	$\frac{3282005975}{2322432}$	$1.413176349189126\mathrm{e}+03$
9	9	$\frac{81540306287}{185794560}$	$4.388734863227427\mathrm{e}+02$
	11	$-\frac{12107862068291}{7431782400}$	$-1.629200293632252\mathrm{e}+03$
	13	$\frac{155601332826841}{59454259200}$	$2.617160400626790\mathrm{e}+03$
10	10	$\frac{61773797}{89600}$	$6.894396986607143\mathrm{e}+02$
	12	$-\frac{536701569}{197120}$	$-2.722714940137987\mathrm{e}+03$
	14	$\frac{37258256823}{7884800}$	$4.725326808923498\mathrm{e}+03$
11	11	$\frac{87445310300933}{81749606400}$	$1.069672554422635\mathrm{e}+03$
	13	$-\frac{17578844792644837}{3923981107200}$	$-4.479849497845420\mathrm{e}+03$
	15	$\frac{262132030525828801}{31391848857600}$	$8.350321502722398\mathrm{e}+03$
12	12	$\frac{449524674151}{273715200}$	$1.642308041902678\mathrm{e}+03$
	14	$-\frac{2070332580533}{284663808}$	$-7.272904114783008\mathrm{e}+03$
	16	$\frac{824780056991401}{56932761600}$	$1.448691463073875\mathrm{e}+04$
13	13	$\frac{20984404436665}{8396996608}$	$2.499036907633463\mathrm{e}+03$
	15	$-\frac{5487944414121525}{470231810048}$	$-1.167072132691604\mathrm{e}+04$
	17	$\frac{7157654925694575}{289373421568}$	$2.473501155327285\mathrm{e}+04$
14	14	$\frac{328959702110873}{87178291200}$	$3.773413054818778\mathrm{e}+03$
	16	$-\frac{6060295438505719}{326918592000}$	$-1.853762859258160\mathrm{e}+04$
	18	$\frac{108918830087067769}{2615348736000}$	$4.164600635770348\mathrm{e}+04$
15	15	$\frac{4949043858894383777}{87487218176000}$	$5.659366719180949\mathrm{e}+03$
	17	$-\frac{1454980745516583126743}{49861671203635200}$	$-2.918034454911144\mathrm{e}+04$
	19	$\frac{151945549173974447638099 3}{21939135329599488000}$	$6.925776558248173\mathrm{e}+04$
16	16	$\frac{37256589643389}{4415488000}$	$8.437706011971723\mathrm{e}+03$
	18	$-\frac{88927518311515449}{1951645696000}$	$-4.556540077626643\mathrm{e}+04$
	20	$\frac{1778602718490427737}{15613165568000}$	$1.139168550249520\mathrm{e}+05$
17	17	$\frac{44877721528232900237939}{3586204813492224000}$	$1.251398731031518\mathrm{e}+04$
	19	$-\frac{47423264674413886530246479}{671337541085744332800}$	$-7.063997135884422\mathrm{e}+04$
	21	$\frac{49823033163906229933959395 29}{26853501643429773312000}$	$1.855364481901614\mathrm{e}+05$
18	18	$\frac{1892289988154335435}{102437979291648}$	$1.847254310597884\mathrm{e}+04$
	20	$-\frac{211767548874661866325}{1946321606541312}$	$-1.088039860231429\mathrm{e}+05$
	22	$\frac{423962129874415950925}{1415506622939136}$	$2.995126430380860\mathrm{e}+05$

q	n	c_n (exact)	c_n (decimal)
19	19	$\frac{316865097671621393629}{11669426143232000}$	$2.715344300417000\mathrm{e}+04$
	21	$-\frac{26440607335165997253220 29}{15870419554795520000}$	$-1.666030771516467\mathrm{e}+05$
	23	$\frac{60895737329297676977470929}{126963356438364160000}$	$4.796323840009713\mathrm{e}+05$
20	20	$\frac{35177078028907807663759}{884691639336960000}$	$3.976196503368290\mathrm{e}+04$
	22	$-\frac{145051523305648303817561}{571646905417728000}$	$-2.537432144404817\mathrm{e}+05$
	24	$\frac{22664909709637753062285 9757}{297256390817218560000}$	$7.624700564831351\mathrm{e}+05$
21	21	$\frac{62170196188259552480487 93086027}{10714547155728479551488 0000}$	$5.802410058461553\mathrm{e}+04$
	23	$-\frac{36260626911224353323028 55058472703}{94288014970410620055094 40000}$	$-3.845730225904494\mathrm{e}+05$
	25	$\frac{90819146776613122776836623 696089409}{75430411976328496042475520000}$	$1.204012339281852\mathrm{e}+06$
22	22	$\frac{1377166963898731019689}{16315758018560000}$	$8.440717019289780\mathrm{e}+04$
	24	$-\frac{15242084837812958699 6319}{262683704098816000}$	$-5.802447810801088\mathrm{e}+05$
	26	$\frac{19854779923346362963900077}{10507348163952640000}$	$1.889609025373479\mathrm{e}+06$
23	23	$\frac{42481594433570398023878491 648025}{34697989509111081795387392}$	$1.224324378287550\mathrm{e}+05$
	25	$-\frac{17083335493495729260989833 13335625}{19594158781615684619033640 96}$	$-8.718585821364402\mathrm{e}+05$
	27	$\frac{78581793010463827708227572 5288200625}{26648055942997331081885 7517056}$	$2.948875264242825\mathrm{e}+06$
24	24	$\frac{19890013326419723065837 21133}{11229835325488496640000}$	$1.771175867670572\mathrm{e}+05$
	26	$-\frac{12456965645939194860409 453115969}{95453600266652221440000 00}$	$-1.305028370971899\mathrm{e}+06$
	28	$\frac{20563835676992918590595 915774057}{44919341301953986560000 00}$	$4.577946844491772\mathrm{e}+06$

Table 2: Exact coefficients c_n of $G_{21q}(e) = \sum_n c_n e^n$ (terms printed in Table C.4 of the main document).

q	n	c_n (exact)	c_n (decimal)
-24	24	$\frac{166006512122547715717303}{42029392655810560000}$	$3.949771853284140\mathrm{e}+03$
	26	$-\frac{42004229947296156955039959}{2101469632790528000000}$	$-1.998802613746029\mathrm{e}+04$
	28	$\frac{54356045524426175066281 15011}{10927642090510745600000 0}$	$4.974178791198462\mathrm{e}+04$
-23	23	$\frac{99855990709256012500492 2034579}{34921487026078007427072 0000}$	$2.859442687382912\mathrm{e}+03$
	25	$-\frac{59242324897719753654403 4209702447}{43103092557901997738557 440000}$	$-1.374433280353082\mathrm{e}+04$
	27	$\frac{65427773312934743282344 5216723460357}{20114776527020932277993 472000000}$	$3.252721859725519\mathrm{e}+04$
-22	22	$\frac{51247759234761110624927}{24771365901434880000}$	$2.068830577961492\mathrm{e}+03$
	24	$-\frac{44716798705772835553332 1}{47478451311083520000}$	$-9.418335575603327\mathrm{e}+03$
	26	$\frac{19275347634295609578081 8909}{91158626512803584000 0}$	$2.114484209636672\mathrm{e}+04$
-21	21	$\frac{2158125580837927625237}{1442765414072320000}$	$1.495825696809883\mathrm{e}+03$

q	n	c_n (exact)	c_n (decimal)
	23	$-\frac{897892766865346204043217}{1396596920822005760000}$	$-6.429147546437773\mathrm{e}+03$
	25	$\frac{3509304059180339563095110787}{256973833431249059840000}$	$1.365627002688280\mathrm{e}+04$
-20	20	$\frac{1869766015004508449}{1730063650258944}$	$1.080749841038886\mathrm{e}+03$
	22	$-\frac{86596982965180747819}{19817092721147904}$	$-4.369812675537839\mathrm{e}+03$
	24	$\frac{13997898736651036292975}{1598578812839264256}$	$8.756464569794416\mathrm{e}+03$
-19	19	$\frac{873010514859629520105271}{1118895901809573888000}$	$7.802428388983483\mathrm{e}+02$
	21	$-\frac{9798989160038875311878743}{3315247116472811520000}$	$-2.955734162726400\mathrm{e}+03$
	23	$\frac{93066538146554032366662780979}{1670884546702297006080000}$	$5.569896395907945\mathrm{e}+03$
-18	18	$\frac{28530109089289}{50692096000}$	$5.628117860679700\mathrm{e}+02$
	20	$-\frac{147454891059111513}{74162536448000}$	$-1.988266557772082\mathrm{e}+03$
	22	$\frac{20833375547140925061}{5933002915840000}$	$3.511438615935910\mathrm{e}+03$
-17	17	$\frac{123587915320026102523}{304710212911104000}$	$4.055916411179876\mathrm{e}+02$
	19	$-\frac{87474306152163229889671}{65817405988798464000}$	$-1.329045179432483\mathrm{e}+03$
	21	$\frac{812077886897622837537007}{370527618899902464000}$	$2.191679770886403\mathrm{e}+03$
-16	16	$\frac{4072803723077897}{13948526592000}$	$2.919880961057063\mathrm{e}+02$
	18	$-\frac{19916471928549961}{22583328768000}$	$-8.819103743807287\mathrm{e}+02$
	20	$\frac{7697920953778174313}{5690998849536000}$	$1.352648481804877\mathrm{e}+03$
-15	15	$\frac{8975566512641}{42748346368}$	$2.099628938947639\mathrm{e}+02$
	17	$-\frac{17460955803288969}{30094835843072}$	$-5.801977420424633\mathrm{e}+02$
	19	$\frac{3373819792467327855}{4092897674657792}$	$8.243108087840026\mathrm{e}+02$
-14	14	$\frac{8832007763}{58572800}$	$1.507868458226344\mathrm{e}+02$
	16	$-\frac{13443781723441}{35582976000}$	$-3.778149900514504\mathrm{e}+02$
	18	$\frac{187961928302173}{379551744000}$	$4.952208263392224\mathrm{e}+02$
-13	13	$\frac{141437610652069}{1307993702400}$	$1.081332504832013\mathrm{e}+02$
	15	$-\frac{5932754076534017}{24415882444800}$	$-2.429874934869516\mathrm{e}+02$
	17	$\frac{857842182580542163}{2929905893376000}$	$2.927883057677627\mathrm{e}+02$
-12	12	$\frac{610433321}{7884800}$	$7.741899870637175\mathrm{e}+01$
	14	$-\frac{31546570047}{205004800}$	$-1.538821044531640\mathrm{e}+02$
	16	$\frac{244045001913}{1435033600}$	$1.700622214789953\mathrm{e}+02$
-11	11	$\frac{6526503251}{117964800}$	$5.532585356818305\mathrm{e}+01$
	13	$-\frac{11361822472981}{118908518400}$	$-9.555095485052314\mathrm{e}+01$
	15	$\frac{399499342349057}{4122161971200}$	$9.691500361708471\mathrm{e}+01$

q	n	c_n (exact)	c_n (decimal)
-10	10	$\frac{7635529}{193536}$	$3.945275814318783\mathrm{e}+01$
	12	$-\frac{13672591}{236544}$	$-5.780147033955627\mathrm{e}+01$
	14	$\frac{68362955}{1261568}$	$5.418887844333401\mathrm{e}+01$
-9	9	$\frac{64370707}{2293760}$	$2.806340114048549\mathrm{e}+01$
	11	$-\frac{441911811}{13107200}$	$-3.371519554138184\mathrm{e}+01$
	13	$\frac{240836615967}{8074035200}$	$2.982853183089913\mathrm{e}+01$
-8	8	$\frac{178331}{8960}$	$1.990301339285714\mathrm{e}+01$
	10	$-\frac{9009241}{483840}$	$-1.862028976521164\mathrm{e}+01$
	12	$\frac{17569129}{1075200}$	$1.634033575148809\mathrm{e}+01$
-7	7	$\frac{432091}{30720}$	$1.406546223958333\mathrm{e}+01$
	9	$-\frac{3073843}{327680}$	$-9.380624389648437\mathrm{e}+00$
	11	$\frac{215487317}{23592960}$	$9.133543099297418\mathrm{e}+00$
-6	6	$\frac{3167}{320}$	$9.896875000000000\mathrm{e}+00$
	8	$-\frac{2187}{560}$	$-3.905357142857143\mathrm{e}+00$
	10	$\frac{195147}{35840}$	$5.444949776785714\mathrm{e}+00$
-5	5	$\frac{1773}{256}$	$6.925781250000000\mathrm{e}+00$
	7	$-\frac{4987}{6144}$	$-8.116861979166666\mathrm{e}-01$
	9	$\frac{418315}{114688}$	$3.647417340959822\mathrm{e}+00$
-4	4	$\frac{77}{16}$	$4.812500000000000\mathrm{e}+00$
	6	$\frac{129}{160}$	$8.062500000000000\mathrm{e}-01$
	8	$\frac{899}{320}$	$2.809375000000000\mathrm{e}+00$
	10	$\frac{107279}{40320}$	$2.660689484126984\mathrm{e}+00$
-3	3	$\frac{53}{16}$	$3.312500000000000\mathrm{e}+00$
	5	$\frac{393}{256}$	$1.535156250000000\mathrm{e}+00$
	7	$\frac{24753}{10240}$	$2.417285156250000\mathrm{e}+00$
	9	$\frac{211557}{81920}$	$2.582482910156250\mathrm{e}+00$
-2	2	$\frac{9}{4}$	$2.250000000000000\mathrm{e}+00$
	4	$\frac{7}{4}$	$1.750000000000000\mathrm{e}+00$
	6	$\frac{141}{64}$	$2.203125000000000\mathrm{e}+00$
	8	$\frac{197}{80}$	$2.462500000000000\mathrm{e}+00$
	10	$\frac{62401}{23040}$	$2.708376736111111\mathrm{e}+00$
-1	1	$\frac{3}{2}$	$1.500000000000000\mathrm{e}+00$

q	n	c_n (exact)	c_n (decimal)
	3	$\frac{27}{16}$	1.687500000000000e + 00
	5	$\frac{261}{128}$	2.039062500000000e + 00
	7	$\frac{14309}{6144}$	2.328938802083333e + 00
	9	$\frac{423907}{163840}$	2.587322998046875e + 00
0	0	1	1.000000000000000e + 00
	2	$\frac{3}{2}$	1.500000000000000e + 00
	4	$\frac{15}{8}$	1.875000000000000e + 00
	6	$\frac{35}{16}$	2.187500000000000e + 00
	8	$\frac{315}{128}$	2.460937500000000e + 00
	10	$\frac{693}{256}$	2.707031250000000e + 00